

一. 计算下列各题.

1. 求极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{\sin(x^2 y^3)}{\ln(1+x^4+y^4)}$.

方法一: 原式 = $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^3}{x^4+y^4} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^2}{x^4+y^4} \cdot y$
 $= 0. \quad \left(\left| \frac{x^2 y^2}{x^4+y^4} \right| \leq \frac{1}{2} \right)$

方法二: 令 $x^2 = \rho \cos \theta, y^2 = \rho \sin \theta$

原式 = $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^3}{x^4+y^4} = \lim_{\rho \rightarrow 0^+} \sqrt{\rho} \cdot \sin \theta \cos \theta \sqrt{\sin \theta}$
 $= 0.$

方法三: $0 \leq \left| \frac{x^2 y^3}{x^4+y^4} \right| \leq \frac{|y|}{2}$

而 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} 0 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{|y|}{2} = 0$

∴ 由夹逼原理 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^3}{x^4+y^4} = 0.$

2. 设 $z = f(x^2 y^3, e^{xy})$, 其中 f 具有二阶连续偏导数. 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

解: $\frac{\partial z}{\partial x} = 2x f_1' + y e^{xy} f_2'$

$\frac{\partial z}{\partial y} = -2y f_1' + x e^{xy} f_2'$

3. 设由 $F(x, y, \frac{z}{y}) = 0$ 确定 $z = z(x, y)$.

$F(1, 0)$ 二阶连续可微且 $F_0' \neq 0$. 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

解: $\frac{\partial z}{\partial x} = -\frac{F_x'}{F_z'} = -\frac{y F_1'}{y F_2'} = -\frac{y^2 F_1'}{F_2'}$

$\frac{\partial z}{\partial y} = -\frac{F_y'}{F_z'} = -\frac{x F_1' - \frac{z}{y^2} F_2'}{y F_2'}$
 $= -\frac{xy F_1'}{F_2'} + \frac{z}{y}$

4. 求曲面 $\sqrt{x} + 2\sqrt{y} + 3\sqrt{z} = 6$ 在 $(1, 1, 1)$ 处的切平面.

解: 记 $F(x, y, z) = \sqrt{x} + 2\sqrt{y} + 3\sqrt{z} - 6$.

$\vec{n}_{(1,1,1)} = (F_x', F_y', F_z')|_{(1,1,1)}$

$= \left(\frac{1}{2\sqrt{x}}, \frac{1}{\sqrt{y}}, \frac{3}{2\sqrt{z}} \right)|_{(1,1,1)}$

$= \left(\frac{1}{2}, 1, \frac{3}{2} \right) = \frac{1}{2} (1, 2, 3)$

∴ 切平面为 $(x-1) + 2(y-1) + 3(z-1) = 0$

即 $x + 2y + 3z - 6 = 0$.

5. 求函数 $f(x, y) = x^4 + y^4 - (x+y)^2$

的所有驻点, 并判断是否取得极值.

解: $\begin{cases} f_x' = 4x^3 - 2(x+y) = 0 \\ f_y' = 4y^3 - 2(x+y) = 0 \end{cases}$ 解得驻点 $(1, 1), (-1, -1), (0, 0)$.

$f_{xx}'' = 12x^2 - 2, f_{xy}'' = -2, f_{yy}'' = 12y^2 - 2$

在 $(1, 1)$ 和 $(-1, -1)$ 处, $A=10, B=-2, C=10$,

$B^2 - AC < 0, A > 0$, ∴ 取极小值.

在 $(0, 0)$ 处, $A=B=C=-2, B^2 - AC = 0$.

$\forall \varepsilon > 0, (\varepsilon < 1) \quad f(\varepsilon, -\varepsilon) = 2\varepsilon^4 > 0$

$f(\varepsilon, 0) = \varepsilon^2(\varepsilon^2 - 1) < 0$

$f(0, 0) = 0$

∴ 在 $(0, 0)$ 点的任意邻域内, 都有点比 $(0, 0)$ 点的函数值大, 也有点比 $(0, 0)$ 点的函数值小, ∴ $(0, 0)$ 不是极值点.

6. 函数 $f(x, y, z) = xyz + \ln(xyz)$ 在点 $(2, 1, 1)$ 处沿什么方向的方向导数取得最大值?

解: 沿梯度方向方向导数取得最大值.

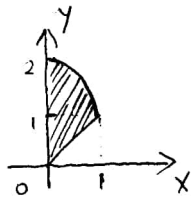
$\text{grad } f = (f_x', f_y', f_z')|_{(2,1,1)}$

$= \left(yz + \frac{1}{x}, zx + \frac{1}{y}, xy + \frac{1}{z} \right)|_{(2,1,1)} = \left(\frac{3}{2}, 3, 3 \right)$

∴ 沿 $\left(\frac{3}{2}, 3, 3 \right)$ 方向, 方向导数取最大值.



7. 交换累次积分 $I_1 = \int_0^1 dx \int_x^{1+\sqrt{1-x^2}} f(x,y) dy$ 的顺序.



$$I_1 = \int_0^1 dy \int_0^y f(x,y) dx + \int_1^2 dy \int_0^{\sqrt{2y-y^2}} f(x,y) dx.$$

8. 求二重积分 $I_2 = \iint_D (|x|+|y|) dx dy$.

其中 $D = \{(x,y) | x^2+y^2 \leq 1\}$.

解: 设 D_1 是 D 中第一象限的部分.

则由对称性

$$I_2 = 4 \iint_{D_1} (x+y) dx dy \quad \begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases}$$

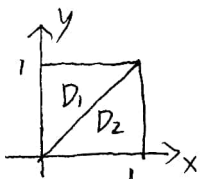
$$= 4 \int_0^{\frac{\pi}{2}} d\theta \int_0^1 \rho^2 (\cos \theta + \sin \theta) d\rho = \frac{8}{3}.$$

9. 设 $f(x)$ 在 $[0,1]$ 上连续, 并设 $\int_0^1 f(x) dx = A$.

求 $I_3 = \int_0^1 dx \int_x^1 f(x)f(y) dy$.

方法一: $I_3 = \int_0^1 dx \int_x^1 f(x)f(y) dy$

$$= \iint_{D_1} f(x)f(y) dx dy \quad ①$$



又由轮换对称性

$$I_3 = \int_0^1 dy \int_y^1 f(y)f(x) dx$$

$$= \iint_{D_2} f(x)f(y) dx dy \quad ②$$

$$①+② \Rightarrow 2I_3 = \iint_{D_1+D_2} f(x)f(y) dx dy$$

$$= \int_0^1 f(x) dx \cdot \int_0^1 f(y) dy = A^2$$

$$\therefore I_3 = \frac{A^2}{2}.$$

方法2. 由轮换对称性

$$I_3 = \int_0^1 dy \int_y^1 f(y)f(x) dx \quad ①$$

又 $I_3 = \int_0^1 dx \int_x^1 f(x)f(y) dy$ 交换积分次序

$$= \int_0^1 dy \int_0^y f(x)f(y) dx \quad ②$$

$$①+② \Rightarrow 2I_3 = \int_0^1 dy \int_0^1 f(x)f(y) dx$$

$$= \int_0^1 f(y) dy \cdot \int_0^1 f(x) dx = A^2$$

$$\therefore I_3 = \frac{A^2}{2}.$$

方法3. 交换积分次序

$$I_3 = \int_0^1 dy \int_0^y f(x)f(y) dx$$

$$= \int_0^1 f(y) dy \int_0^y f(x) dx.$$

$$\left(\int_0^y f(x) dx = F(y) \right) = \int_0^1 f(y) F(y) dy$$

$$②) F'(y) = f(y). \quad = \int_0^1 F(y) dF(y)$$

$$= \frac{F^2(y)}{2} \Big|_0^1 = \frac{F^2(1) - F^2(0)}{2} = \frac{A^2}{2}.$$

10. 求第一类曲线积分 $I_4 = \oint_C (x+y) ds$.

其中 C 是双纽线 $(x^2+y^2)^2 = 2(x^2-y^2)$ 的右半支.

解: 双纽线的极坐标方程 $\rho = \sqrt{2 \cos 2\theta}$.

$$\therefore \text{参数方程为} \begin{cases} x = \sqrt{2 \cos 2\theta} \cos \theta \\ y = \sqrt{2 \cos 2\theta} \sin \theta \end{cases}$$

C 关于 $y=0$ 对称, y 关于 y 是奇函数,

x 关于 y 是偶函数. 设 C_1 是 C 中 $y \geq 0$ 的部分.

$$②) I_4 = 2 \int_{C_1} x ds$$

$$= 2 \int_0^{\frac{\pi}{4}} \sqrt{2 \cos 2\theta} \cdot \cos \theta \cdot \sqrt{\rho'^2(\theta) + \rho^2(\theta)} d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} \cos \theta d\theta = 2\sqrt{2}$$



11. 求 $I_5 = \oint_C (y-z)dx + (z-x)dy + (x-y)dz$.

其中 C 为球面 $x^2+y^2+z^2=1$ 与 $y=x\tan\beta$ ($0<\beta<\frac{\pi}{2}$) 的交线. 从 x 轴正向看去沿逆时针方向.

解: $\begin{cases} x^2+y^2+z^2=1 \\ y=x\tan\beta \end{cases} \Rightarrow (1+\tan^2\beta)x^2+z^2=1$

即 $\frac{x^2}{\cos^2\beta} + z^2 = 1$

设 C 的参数方程为 $\begin{cases} x = \cos\beta \cos\theta \\ y = x\tan\beta = \sin\beta \cos\theta \\ z = \sin\theta \end{cases}$

则 $I_5 = \int_0^{2\pi} [(\sin\beta \cos\theta - \sin\theta)(-\cos\beta \sin\theta) + (\sin\theta - \cos\beta \cos\theta)(-\sin\beta \sin\theta) + (\cos\beta \cos\theta - \sin\beta \cos\theta) \cdot \cos\theta] d\theta$
 $= \int_0^{2\pi} (\cos\beta - \sin\beta) d\theta = 2\pi(\cos\beta - \sin\beta)$.

12. 证明: $(2x\cos y + y^2\cos x)dx + (2y\sin x - x^2\sin y)dy$

在 xOy 平面上是某个函数的全微分. 并求出一个原函数.

解: 设 $p = 2x\cos y + y^2\cos x$, $Q = 2y\sin x - x^2\sin y$

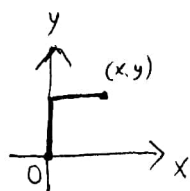
则 $\frac{\partial Q}{\partial x} = \frac{\partial p}{\partial y} = 2y\cos x - 2x\sin y$.

$\therefore pdx + Qdy$ 是恰当微分.

$u(x, y) = \int_{(0,0)}^{(x,y)} pdx + Qdy$

$= \int_0^y 0 dy + \int_0^x (2x\cos y + y^2\cos x) dx$

$= x^2\cos y + y^2\sin x$.



二. 讨论函数

$f(x, y) = \begin{cases} \frac{(x+y)\ln(1+xy)}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

在 $(0, 0)$ 处的连续性, 可偏导性及可微性.

解: $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{(x+y)\ln(1+xy)}{x^2+y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2} (x+y)$

$\begin{pmatrix} x = \rho \cos\theta \\ y = \rho \sin\theta \end{pmatrix} = \lim_{\rho \rightarrow 0} \rho(\cos\theta + \sin\theta) \cos\theta \sin\theta = 0$
 $= f(0, 0)$, $\therefore$ 连续.

$f'_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0}{x} = 0$.

$f'_y(0, 0) = \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0}{y} = 0$.

$\therefore f(x, y)$ 在 $(0, 0)$ 处可偏导.

$\Delta f(x, y) = f(x, y) - f(0, 0)$
 $= f'_x(0, 0)x + f'_y(0, 0)y + \omega$

$\Rightarrow \omega = f(x, y) = \frac{(x+y)\ln(1+xy)}{x^2+y^2}$

$\lim_{\rho \rightarrow 0} \frac{\omega}{\rho} = \lim_{\rho \rightarrow 0} \frac{(x+y)\ln(1+xy)}{(x^2+y^2)\sqrt{x^2+y^2}}$

$= \lim_{\rho \rightarrow 0} \frac{(x+y)xy}{(x^2+y^2)^{\frac{3}{2}}} \begin{pmatrix} x = \rho \cos\theta \\ y = \rho \sin\theta \end{pmatrix}$

$= \lim_{\rho \rightarrow 0} (\cos\theta + \sin\theta) \cos\theta \sin\theta$

$\neq 0$, $\therefore \omega \neq o(\rho)$

$\therefore f(x, y)$ 在 $(0, 0)$ 处不可微.

三. 在椭圆抛物面 $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ ($a>0, b>0$)

及平面 $z=1$ 所围立体内嵌入一个长方体.

且有一面在 $z=1$ 上. 求此长方体体积的

最大值.



解：设长方体在第一卦限的顶点为 (x, y, z)

$$|2| \cdot \frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0.$$

$$V = 2x \cdot 2y \cdot (1-z) = 4xy(1-z).$$

$$\text{设 } F(x, y, z, \lambda) = 4xy(1-z) + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - z \right).$$

$$|2| \begin{cases} F'_x = 4y(1-z) + \frac{2x\lambda}{a^2} = 0 \\ F'_y = 4x(1-z) + \frac{2y\lambda}{b^2} = 0 \\ F'_z = -4xy - \lambda = 0 \\ F'_\lambda = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0 \end{cases}$$

$$\text{解得 } x = \frac{a}{2}, y = \frac{b}{2}, z = \frac{1}{2}.$$

$$\therefore V_{\max} = \frac{ab}{2}.$$

方法2. 设长方体在第一卦限顶点为 (x, y, z) . $|2|$

$$x = a\sqrt{z}\cos\theta, y = b\sqrt{z}\sin\theta.$$

$$V = 2x \cdot 2y \cdot (1-z) = 2ab(z-z^2)\sin 2\theta.$$

$$\begin{cases} V'_x = 2ab\sin 2\theta(1-2z) = 0 \\ V'_y = 2ab(z-z^2)2\cos 2\theta = 0 \end{cases}$$

$$\text{解得 } z = \frac{1}{2}, \theta = \frac{\pi}{4}$$

$$\therefore V_{\max} = \frac{ab}{2}.$$

四. 计算 $I_6 = \iiint_{\Omega} (y+z)^2 dx dy dz$, 其中 Ω 为

$$\text{球体 } x^2 + y^2 + z^2 \leq 2x.$$

$$\text{解法一. } I_6 = \iiint_{\Omega} (y^2 + z^2) dx dy dz$$

$$= 2 \iiint_{\Omega} z^2 dx dy dz \quad (\text{对称性})$$

$$= 2 \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^{1+\cos\varphi} r^2 \sin\varphi dr \quad \begin{matrix} x = 1+r\cos\varphi \\ y = r\sin\varphi\cos\theta \\ z = r\sin\varphi\sin\theta \end{matrix}$$

$$= \frac{8}{15}\pi.$$

$$\text{解法二. } (x-1)^2 + y^2 + z^2 \leq 1$$

$$\text{设 } x-1=u, y=v, z=w.$$

$$|2| \Omega': u^2 + v^2 + w^2 \leq 1.$$

$$I_6 = \iiint_{\Omega} (v^2 + w^2) du dv dw$$

$$= \frac{2}{3} \iiint_{\Omega'} (u^2 + v^2 + w^2) du dv dw \quad (\text{轮换对称性})$$

$$\begin{pmatrix} u = r\sin\varphi\cos\theta \\ v = r\sin\varphi\sin\theta \\ w = r\cos\varphi \end{pmatrix} = \frac{2}{3} \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^1 r^2 \cdot r^2 \sin\varphi dr = \frac{8}{15}\pi.$$

$$\text{解法三. } I_6 = \iiint_{\Omega} (y^2 + z^2) dx dy dz$$

$$\begin{pmatrix} x = 1+r\cos\varphi \\ y = r\sin\varphi\cos\theta \\ z = r\sin\varphi\sin\theta \end{pmatrix} = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^1 r^2 \sin^2\varphi \cdot r^2 \sin\varphi dr = \frac{8}{15}\pi.$$

五. 求曲面 $(2x+3y)^2 + (2y+3z)^2 + (2z+3x)^2 = 1$ 所围之长方体体积.

解：设 $2x+3y=u, 2y+3z=v, 2z+3x=w$.

$$|2| \Omega': u^2 + v^2 + w^2 \leq 1.$$

$$J(u, v, w) = \frac{D(x, y, z)}{D(u, v, w)} = \frac{1}{D(x, y, z)}$$

$$= \frac{1}{\begin{vmatrix} 2 & 3 & 0 \\ 0 & 2 & 3 \\ 3 & 0 & 2 \end{vmatrix}} = \frac{1}{35}$$

$$\therefore V = \iiint_{\Omega} dx dy dz = \iiint_{\Omega'} \frac{1}{35} du dv dw = \frac{1}{35} \cdot \frac{4}{3}\pi = \frac{4}{105}\pi.$$



六. 设 D 为两条直线 $y=x$, $y=4x$ 和
 两条曲线 $xy=1$, $xy=4$ 所围成的闭区域
 (在第一象限的部分). $F(u)$ 连续可微,
 C 是 D 的边界曲线, 取正向.

记 $f(u) = F'(u)$. 证明:

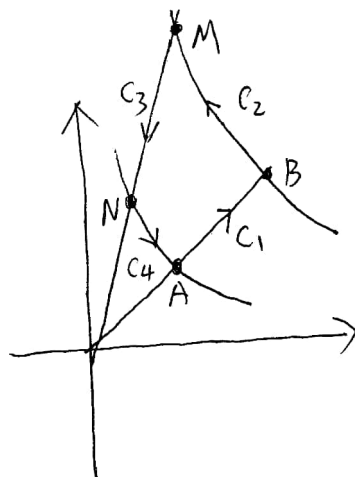
$$I_7 = \int_C \frac{F(xy)}{y} dy = \ln 2 \int_1^4 f(u) du.$$

方法一. 由格林公式

$$\begin{aligned} I_7 &= \int_C \frac{F(xy)}{y} dy = \iint_D \frac{\partial}{\partial x} \left(\frac{F(xy)}{y} \right) dx dy \\ &= \iint_D f(xy) dx dy \end{aligned}$$

$$\begin{aligned} xy &= u, \\ \frac{y}{x} &= v, \\ J(u, v) &= \frac{1}{\begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{2v} \end{aligned}$$

$$\begin{aligned} &= \iint_{D'} f(u) \cdot \frac{1}{2v} du dv \\ &= \int_1^4 du \int_1^4 \frac{f(u)}{2v} dv \\ &= \ln 2 \int_1^4 f(u) du. \end{aligned}$$



$A(1, 1)$
 $B(2, 2)$
 $M(1, 4)$
 $N(\frac{1}{2}, 2)$

方法二:

$$\begin{aligned} I_7 &= \int_C \frac{F(xy)}{y} dy \\ &= \int_{C_1+C_2+C_3+C_4} \frac{F(xy)}{y} dy \\ &= \int_1^2 \frac{F(y^2)}{y} dy + \int_2^4 \frac{F(4)}{y} dy \\ &\quad + \int_4^1 \frac{F(\frac{y^2}{4})}{y} dy + \int_2^1 \frac{F(1)}{y} dy \\ &\quad \text{对于 } \int_4^1 \frac{F(\frac{y^2}{4})}{y} dy \quad \text{令 } \frac{y}{2} = t \\ &= \int_2^1 \frac{F(t^2)}{t} dt = - \int_1^2 \frac{F(y^2)}{y} dy \\ \therefore I_7 &= F(4) \int_2^4 \frac{1}{y} dy + F(1) \int_2^1 \frac{1}{y} dy \\ &= \ln 2 (F(4) - F(1)) \\ &= \ln 2 \int_1^4 f(u) du. \end{aligned}$$

